LOW LEVEL DESIGN DOCUMENT

Real-Time Luxury Auction Engine

Core Engine, Bidding Lua & API Implementation

PT Ako Media Asia (SALT)

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Core Engine, Bidding Lua & API Implementation

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1. Governance

1.1 Contributors

Name	Position
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1.2 Approvals

Name	Position	Date	Signature
Geri Setyo Andrian	Project Manager Delivery Team	24 Oct 2026	`[ APPROVED & SIGNED ]`
Alini	Technical Architect / Tech Lead Delivery Team	24 Oct 2026	`[ APPROVED & SIGNED ]`
Daniel H. S.	System Analyst Delivery Team	24 Oct 2026	`[ APPROVED & SIGNED ]`

1.3 Revision History

Version	Date	Author	Summary of Changes
0.1	24 Oct 2026	Daniel H. S.	Initial Technical Draft (Components & Goroutines)
0.5	24 Oct 2026	Daniel H. S.	Added Lua Scripting Logic, Settlement Daemon, and 14 API Endpoints
1.0	24 Oct 2026	Daniel H. S.	Baseline Alignment with Enterprise Template Structure & Sign-Off

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3. Introduction

3.1 Low-Level Design Overview

Dokumen Low-Level Design (LLD) ini menguraikan spesifikasi implementasi perangkat lunak pada tingkat kode, skrip in-memory, antarmuka REST API, WebSocket pub/sub, dan state machine database untuk **Real-Time Luxury Auction Engine**.

3.2 Glossaries

No	Terms	Expansion	Definition
1	**EVALSHA**	Evaluate SHA1 Digest	Perintah Redis untuk mengeksekusi skrip Lua yang telah di-cache server secara atomic.
2	**BYTEA**	Byte Array	Tipe data kolom PostgreSQL untuk menyimpan representasi biner gambar produk secara langsung.
3	**ZSET**	Sorted Set	Struktur data Redis berurutan berdasarkan score waktu untuk tracking expiry lelang.
4	**Base64**	Binary-to-Text Encoding	Skema representasi string data biner untuk transfer respons JSON ke antarmuka front-end.

3.3 Objective Overview

Menyediakan instruksi teknis langsung bagi tim engineering untuk membangun modul Golang backend, mengonfigurasi skrip Redis Lua, dan menghubungkan Next.js web client sesuai standar kualitas TDD.

3.4 References

- **High-Level Design Document:** [`high-level-design.md`](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/04_Design/high-level-design.md)
- **Database Schema & Redis Lua Design:** [`database-schema.md`](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/06_Data_Design/database-schema.md)
- **API Contracts Specification:** [`api-contracts.md`](file:///d:/Projects/Salt-Hackaton-2026/02_Projects/real-time-auction-engine/05_Integration/api-contracts.md)

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4. System & Integration Landscape

4.1 Component Interaction

Interaksi antar-komponen di dalam modul Golang dan external service:

```
flowchart TD
    subgraph Client ["Client Presentation"]
        NextJS["Next.js Web App"]
    end

    subgraph Transport ["Transport Layer"]
        HTTPHandler["Gin / Chi Router"]
        WSHandler["Gorilla WebSocket Conn Manager"]
    end

    subgraph Usecase ["Business Logic Layer"]
        AuthUC["Auth Usecase"]
        BidUC["Bid Engine Usecase"]
        LotUC["Lot Catalog Usecase"]
        SettleDaemon["Settlement Daemon Goroutine"]
    end

    subgraph Repository ["Data Access Layer"]
        RedisRepo["Redis Hot-Path Client"]
        PGRepo["PostgreSQL GORM / SQLx Repo"]
    end

    NextJS ⟶|HTTP REST| HTTPHandler
    NextJS ⟷|WSS Duplex| WSHandler
    HTTPHandler ⟶ AuthUC
    HTTPHandler ⟶ BidUC
    HTTPHandler ⟶ LotUC
    BidUC ⟶ RedisRepo
    LotUC ⟶ PGRepo
    SettleDaemon ⟶ RedisRepo
    SettleDaemon ⟶ PGRepo
    RedisRepo ⟶|Pub/Sub Event| WSHandler
```

4.2 Impacted Objects & Detailed Functionality

No	Impact	Object Name	Functions
1	**New**	`internal/domain`	Definisi struct murni: `User`, `Wallet`, `Auction`, `Bid`, `LedgerEntry` (seluruh atribut moneter menggunakan `float64`).
2	**New**	`internal/usecase/bid`	Validasi kelipatan tombol bid diskrit, kalkulasi buy-out, pemicu Lua atomik.
3	**New**	`internal/usecase/auth`	Login NIRO SSO, progressive lockout Superadmin, serta **Token Revocation (Auto-Logout)** saat user dipromosikan ke `ADMIN`.
4	**New**	`internal/repository/redis`	Eksekusi EVALSHA `bid_placement.lua`, manajemen connection pool, ZSET expiry index, dan tracking `token_version:{user_id}`.
5	**New**	`internal/transport/ws`	Multiplexing koneksi WebSocket per lot lelang dan proteksi heart-beat ping/pong.

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5. Detailed Sequence Diagrams & Algorithms

5.1 Incremental Bid Execution (bid_placement.lua)

```
sequenceDiagram
    autonumber
    participant UI as Next.js Client
    participant API as Bid Usecase
    participant Lua as Redis (bid_placement.lua)
    participant PS as Redis Pub/Sub
    participant PG as PostgreSQL (bids table)

    UI->>API: POST /api/v1/auctions/{id}/bids {step_increment: 5000000}
    API->>API: Hitung new_price = current_price + step_increment
    API->>Lua: EVALSHA bid_placement.lua (user_id, lot_id, new_price)

    critical Atomic Lua Execution
        Lua->>Lua: Cek lot status == 'ACTIVE' & now < end_time
        Lua->>Lua: Cek wallet available_balance ≥ new_price
        Lua->>Lua: Deduct available_balance, Increase locked_balance
        Lua->>Lua: Update auction current_price & highest_bidder_id
        Lua->>Lua: Evaluasi Anti-Sniping (Velocity ≥ 10 bid/s dalam 2 menit akhir)
        Lua->>Lua: Jika triggered: end_time = end_time + extend_minutes
        Lua->>Lua: Evaluasi Capped Ceiling (jika new_price ≥ ceiling → status = CLOSED_AWARDED)
    end

    Lua-->>API: Return {status: 'OK', new_price, bidder_id, end_time}
    API->>PS: PUBLISH auction:{id}:events {payload JSON}
    API-)PG: Async insert log bid
    API-->>UI: 200 OK {status: 'BID_ACCEPTED'}
```

5.2 Settlement Daemon Expiry Flow

```
sequenceDiagram
    autonumber
    participant Daemon as Settlement Daemon (Ticker 1s)
    participant Redis as Redis ZSET
    participant Lock as Redis Distributed Lock
    participant DB as PostgreSQL (ACID Ledger)

    Daemon->>Redis: ZRANGEBYSCORE auctions:expiry_zset -inf (now)
    Redis-->>Daemon: Daftar lot_id yang kedaluwarsa
    loop Setiap Lot Terpilih
        Daemon->>Lock: SET lock:settle:{lot_id} "locked" NX EX 10
        alt Berhasil Dapatkan Lock
            Daemon->>Redis: HGETALL auction:{lot_id}
            Redis-->>Daemon: Data final pemenang & harga lelang
            Daemon->>DB: BEGIN TRANSACTION (SERIALIZABLE)
            Daemon->>DB: Potong permanen locked_balance pemenang
            Daemon->>DB: Ubah status auction menjadi 'SETTLED'
            Daemon->>DB: Insert ledger_entries 'AUCTION_WIN_PAYMENT'
            Daemon->>DB: Terbitkan kepemilikan di asset_ownership
            Daemon->>DB: Release agunan seluruh penawar yang kalah
            Daemon->>DB: COMMIT TRANSACTION
            Daemon->>Redis: DEL auction:{lot_id} & ZREM auctions:expiry_zset {lot_id}
            Daemon->>Lock: DEL lock:settle:{lot_id}
        else Gagal Lock
            Note over Daemon: Instance lain sedang memproses lot ini
        end
    end
```

5.3 Role Promotion & Mandatory Auto-Logout Sequence

Ketika Superadmin mempromosikan user dari `BIDDER` menjadi `ADMIN` (Auction Manager), sesi lama user wajib dibatalkan seketika agar hak akses lama tidak disalahgunakan.

```
sequenceDiagram
    autonumber
    actor Super as Superadmin Platform
    participant UI as Admin Console
    participant UC as AdminUserUsecase
    participant Redis as Redis Session Cache
    participant DB as PostgreSQL
    actor User as Pengguna (Mantan Bidder)

    Super->>UI: Klik Promosikan ke Auction Manager (ADMIN)
    UI->>UC: PATCH /api/v1/admin/users/{id}/role {role: "ADMIN"}
    UC->>DB: UPDATE users SET role = 'ADMIN', updated_at = NOW() WHERE id = $1
    UC->>Redis: INCR token_version:{user_id}
    Note over Redis: Token JWT lama otomatis INVALID karena token_version tidak cocok
    UC-->>UI: 200 OK {"status": "success", "data": {"new_role": "ADMIN"}}

    par Klien Pengguna Mencoba Akses
        User->>UC: Request API dengan Token JWT Lama
        UC->>Redis: Validasi token_version
        Redis-->>UC: Version Mismatch (Revoked)
        UC-->>User: 401 Unauthorized {"status": "error", "error": {"code": "AUTH_TOKEN_REVOKED", "message":
        User->>User: Auto-Logout & Redirect ke Halaman Login NIRO
    end
```

6. Application Interfaces & API Contracts

6.1 Complete Endpoint Specifications

Berikut adalah 14 endpoint implementasi transport API pada sistem:

Method	Endpoint	Role	Penjelasan Fungsional & Kontrak
`GET`	`/api/v1/auth/niro/authorize-url`	Publik	Mendapatkan redirect URL resmi NIRO SSO beserta CSRF state.
`POST`	`/api/v1/auth/login/niro`	Publik	Menerima `niro_token`. Jika baru, auto-register role `BIDDER`; jika sudah ada, buat session JWT.
`POST`	`/api/v1/auth/login/email`	Publik	Login email & password (min. 12 char) khusus `SUPERADMIN` dengan progressive lockout.
`GET`	`/api/v1/admin/users`	`SUPERADMIN`	Mengambil daftar pengguna terdaftar NIRO (lengkap dengan pagination meta).
`PATCH`	`/api/v1/admin/users/:id/role`	`SUPERADMIN`	Mempromosikan role user dari `BIDDER` menjadi `ADMIN` (Auction Manager).
`GET`	`/api/v1/wallet/me`	`BIDDER`	Mengambil informasi saldo likuid (`available_balance`) dan tertahan (`locked_balance`).
`POST`	`/api/v1/wallet/topup`	`BIDDER`	Menambah saldo e-wallet khusus penawar (minimal deposit Rp 100.000).
`GET`	`/api/v1/auctions/active`	Publik	Daftar lot barang mewah aktif, pagination meta, dan thumbnail Base64 terkompresi.
`GET`	`/api/v1/auctions/:id`	Publik	Detail lengkap lot, foto Base64 utuh, step increment tombol, dan jaminan keaslian.
`POST`	`/api/v1/auctions/:id/bids`	`BIDDER`	Mengirim penawaran tombol bertingkat diskrit; validasi instant buy-out & anti-sniping.
`POST`	`/api/v1/auctions/:id/proxy-bid`	`BIDDER`	Konfigurasi bot penawaran bertingkat dengan batas limit penawaran maksimal.
`DELETE`	`/api/v1/auctions/:id/proxy-bid`	`BIDDER`	Membatalkan konfigurasi bot proxy penawaran pada lot tertentu.
`POST`	`/api/v1/admin/auctions`	`ADMIN`	Membuat lot baru via form biner multipart (`image` disimpan di DB `BYTEA`), durasi extend lot.
`WS`	`/api/v1/ws/auctions`	Universal	Channel WebSocket streaming mutasi harga, countdown timer, dan event lelang.

6.2 WebSocket & Real-Time Event Bus

- **Protokol:** Dupleks penuh WebSocket JSON Frame.
- **Keep-Alive:** Klien mengirim `ping` setiap 30 detik; server menjawab `pong`.
- **Struktur Pesan Broadcast:**

```
{
  "event": "PRICE_UPDATED",
  "data": {
    "lot_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
    "current_price": 125000000,
    "highest_bidder_masked": "Us***78",
    "end_time": "2026-10-25T14:00:00Z",
    "is_extended": true
  }
}
```

---

7. In-Memory, Caching & Data Structures

7.1 In-Memory Atomic Scripts (Lua Engine)

Logika transaksi hot-path dijalankan secara atomik melalui skrip `bid_placement.lua`:

- Validasi status lot lelang (`ACTIVE`).
- Verifikasi saldo `available_balance ≥ target_bid`.
- Pemotongan `available_balance` dan pemindahan dana ke `locked_balance`.
- *Catatan Bisnis:* Dana penawar sebelumnya tetap tersimpan di `locked_balance` hingga lelang mencapai settlement akhir.
- Penghitungan lonjakan velositas penawaran dalam jendela 2 menit terakhir (jika $\ge 10\text{ bid/s}$, perpanjang `end_time`).

7.2 Caching Key Schemas

- `auction:{lot_id}` (Type: `HASH`): Status lot lelang, harga terkini, waktu berakhir, ceiling valuation.
- `wallet:{user_id}` (Type: `HASH`): Saldo bebas (`available_balance`) dan tertahan (`locked_balance`).
- `auctions:expiry_zset` (Type: `ZSET`): Indexing lelang aktif dengan score `end_time` epoch timestamp.
- `lockout:superadmin:{email}` (Type: `STRING`): TTL counter progressive lockout email superadmin.

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8. State Machine & Lifecycle Specifications

```
stateDiagram-v2
    [*] ⟶ DRAFT : AM Create Lot
    DRAFT ⟶ ACTIVE : Scheduled Start Time Reached
    ACTIVE ⟶ CLOSED_AWARDED : Buyout Ceiling Reached
    ACTIVE ⟶ CLOSED_AWARDED : Time-Window Ended with Highest Bidder
    ACTIVE ⟶ EXPIRED_UNSOLD : Time Ended without any Bid
    CLOSED_AWARDED ⟶ SETTLED : Settlement Daemon Debits Winner Funds
    EXPIRED_UNSOLD ⟶ SETTLED : Daemon Releases All Holds
    SETTLED ⟶ [*]
```

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9. Document Sign-Off Section

Dokumen perancangan tingkat rendah (*Low-Level Design*) ini telah ditinjau secara mendalam dari segi kelayakan teknis, struktur data, algoritma usecase, dan kesiapan pengujian TDD, serta secara resmi disahkan untuk tahap koding.

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Tanggal: 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi	**Tanggal:** 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi	**Tanggal:** 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi